

READ LEVA ALALLOŞ

Embedded Software / Firmware Engineer · Konya, Turkey
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Summary

Embedded software engineer building FreeRTOS firmware on STM32 for a commercial fire-safety product — sensor acquisition, point-to-point LoRa links and an AWS IoT Core cloud path — alongside computer-vision and machine-learning work deployed on edge hardware. Comfortable from PCB and firmware through to cloud backend and mobile app.

Skills

Programming: C, C++, Python, Dart

Embedded: STM32 (ARM Cortex-M), ESP32, Raspberry Pi, FreeRTOS, embedded C, UART, I2C, SPI, LoRa (RA-09H), sensor interfacing, fuzzy logic control

Cloud & IoT: AWS IoT Core, AWS Lambda, API Gateway, MQTT, device provisioning, telemetry pipelines

Hardware: PCB design and simulation (Proteus), schematic capture, industrial control panels

AI & Computer Vision: PyTorch, TensorFlow, OpenCV, CNNs, image classification, semantic segmentation, data augmentation, model optimization, edge AI deployment

Robotics & Automation: ROS2, Siemens TIA Portal (PLC, HMI), conveyor automation

Tools: Git, GitHub, STM32CubeIDE, Flutter

Languages: Arabic (Native), Turkish (Fluent), English (Advanced — professional working proficiency)

Experience

İmax Elektronik

June 2026 – Present

Embedded Software Engineer

Konya, Turkey

- Develop **FreeRTOS** firmware in **embedded C** for an **STM32 (ARM Cortex-M)** photoelectric smoke detector, covering sensor acquisition and the **UART, I2C and SPI** peripheral layers.
- Implemented the point-to-point **LoRa** link (Ai-Thinker RA-09H) between detectors and an in-house gateway, removing the need for a LoRaWAN network server.
- Connected the central unit to **AWS IoT Core** over **MQTT** and built the serverless backend behind it with **AWS Lambda** and **API Gateway**.
- Designed a **fuzzy-logic** decision layer that cuts false alarms by separating real fire smoke from cooking smoke, cigarette smoke and steam.
- Designed and simulated the product **PCBs** in **Proteus** and worked on the central unit's Ethernet, **ESP32** Wi-Fi and cellular connectivity.
- Trained a **computer-vision** model for automatic board serial-number recognition on the production line.
- Built the companion **Flutter** application for live alarm monitoring and push notifications, device provisioning and historical sensor reporting.

Selçuk University

Feb 2026 – May 2026

AI & Robotics Intern

Konya, Turkey

- Built and deployed an end-to-end **computer-vision** pipeline on **Raspberry Pi** for automated waste classification across five material categories.
- Integrated the image-recognition software with physical conveyor automation systems and tuned the models for **real-time edge inference**.

Selçuk University

Jun 2025 – Aug 2025

Computer Vision & AI Intern

Konya, Turkey

- Designed and trained **CNN** architectures in **TensorFlow** and **PyTorch** for medical image analysis on clinical MRI data.
- Curated and preprocessed a **2,867-image** MRI dataset, applying **data augmentation** to control model overfitting.
- Benchmarked **precision and recall** against standard medical datasets to validate clinical performance.

Mizan Mekatronik

Jun 2024 – Aug 2024

Automation Department Intern

Konya, Turkey

- Programmed **PLC** systems and designed **HMI** screens in **Siemens TIA Portal**.
- Assembled and wired **industrial control panels** and sensor distribution networks.
- Diagnosed and resolved hardware and software faults during system **commissioning**.

Projects

AI-Powered Automated Waste Sorting System | *Python, OpenCV, ROS2, Raspberry Pi* **Feb 2026 – May 2026**

- Engineered an automated recycling system driven by custom **CNN** models trained on a dataset of **12,000+ images**.
- Optimized the memory footprint for **Raspberry Pi** deployment, reaching **97.69% accuracy** across five waste streams.
- Developed a **ROS2** communication bridge streaming real-time inferences to conveyor control units.

Foldable-Wing Autonomous Drone | *Embedded Systems, Sensor Fusion, Machine Learning* **Aug 2025 – Jan 2026**

- Designed a foldable-wing autonomous **UAV** with a **multi-sensor payload** for real-time air-quality tracking.
- Applied **machine-learning** models to **telemetry** data to predict local environmental pollution patterns.
- Secured **TÜBİTAK 2209-A** undergraduate research funding as principal developer.

Brain Tumor MRI Analysis System | *PyTorch, OpenCV, Deep Learning* **Jun 2025 – Aug 2025**

- Built an end-to-end **deep-learning** platform for automated tumor **classification** and lesion **segmentation**.
- Reached **98.43% peak accuracy** on clinical MRI scans using a **PyTorch** model with **spatial attention**.
- Applied **OpenCV** preprocessing filters to enhance soft-tissue contrast before model training.

Education

Selçuk University, Konya — *MSc in Computer Engineering (ongoing)*, **2026 – Present**

Istanbul University, Istanbul — *Associate Degree in Computer Programming (ongoing)*, **2026 – Present**

Selçuk University, Konya — *BSc in Mechatronics Engineering*, **2022 – 2026**